AD 502 – Machine Learning

Pre-Reading Material – Lecture 36

Basic Ensemble Techniques: Voting Classifiers

Pre-Reading Material – Voting Classifiers

Estimated Reading Time: 5 Minutes

Introduction

Voting Classifiers are an ensemble learning technique used primarily for classification problems. In this approach, multiple machine learning classification models are combined to make a final prediction.

The main idea behind voting classifiers is that combining the decisions of several classifiers can produce more reliable and accurate predictions than relying on a single model.

Different classifiers may learn different patterns from the same dataset. By combining their predictions, voting classifiers attempt to improve overall classification performance and reduce the impact of individual model errors.

Working Principle of Voting Classifiers

Voting classifiers follow a simple process:

1. Select multiple classification algorithms, such as Decision Tree, Logistic Regression, K-Nearest Neighbors, and Support Vector Machine.
2. Train all selected classifiers using the same training dataset.
3. Provide a new input observation to all trained classifiers.
4. Collect the predictions generated by individual classifiers.
5. Combine the predictions using a voting strategy.
6. Assign the final predicted class based on the voting result.

Types of Voting Classifiers

1. Hard Voting

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Hard Voting is a classification technique in which each classifier predicts a class label. The final output is determined by selecting the class that receives the highest number of votes.

For example, consider three classifiers predicting whether an email is Spam or Not Spam:

- Classifier 1: Spam
- Classifier 2: Not Spam
- Classifier 3: Spam

The final prediction is Spam because it receives the majority of votes.

2. Soft Voting

Soft Voting combines the predicted probabilities generated by individual classifiers rather than directly combining their class labels.

Each classifier provides the probability of an input belonging to different classes. These probabilities are combined, and the class with the highest average probability is selected as the final prediction.

Soft Voting generally requires classifiers that can provide probability estimates.

Common Algorithms Used in Voting Classifiers

Some commonly used classification algorithms include:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- K-Nearest Neighbors
- Naive Bayes

These algorithms can be combined to create a voting classifier.

Advantages of Voting Classifiers

1. Improves prediction reliability by combining multiple classifiers.
2. Reduces dependence on a single machine learning model.
3. Can improve classification accuracy.

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4. Combines the strengths of different learning algorithms.
5. Provides flexibility in selecting different base classifiers.

Applications of Voting Classifiers

Voting classifiers are used in several real-world applications:

- Email spam detection.
- Medical diagnosis and disease classification.
- Credit card fraud detection.
- Sentiment analysis.
- Customer churn prediction.
- Image classification.

Conclusion

Voting Classifiers are an important ensemble learning technique that combines predictions from multiple classification models to produce a final decision. Hard Voting uses predicted class labels, whereas Soft Voting uses predicted probabilities. By combining different classifiers, voting techniques can improve the robustness and reliability of machine learning predictions.

Key Terms to Remember: Ensemble Learning, Voting Classifier, Hard Voting, Soft Voting, Base Classifier, Majority Voting, Probability Prediction.

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